Personal Health Digital Twin

Simulate lifestyle changes before you commit to them

CRP Digital Twin Challenge 2026 (Happiest Health)

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Problem

- Chronic lifestyle conditions (hypertension, prediabetes, obesity) build up silently over months
- People get generic advice and cannot see how their own habits change their own numbers
- Check-ups are snapshots; there is no way to test 'what if I sleep more and walk daily?'
- Our question: how can people see in advance how their own habits will change their own health markers?

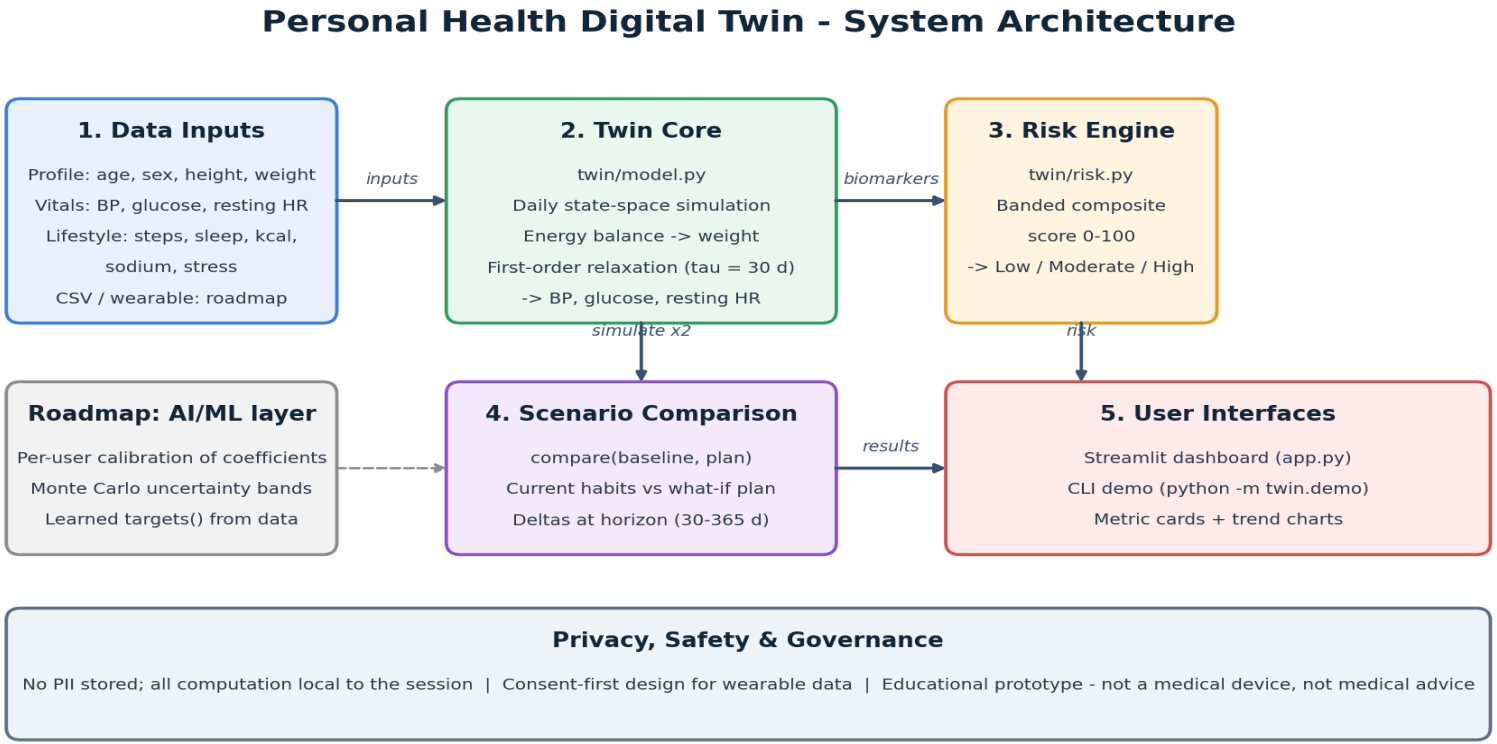
Healthcare Use Case: Preventive Care

- Individual: enter profile and habits, see projected BP, glucose, heart rate, weight and risk
- Compare 'current habits' vs a 'what-if plan' side by side over 30-365 days
- Coach / clinician: use the twin to motivate and personalise lifestyle plans
- Focus: prevention and early behaviour change, not diagnosis

Our Solution

- A virtual copy of the user's key health markers that evolves day by day
- Inputs: steps, sleep, calories, sodium, stress
- Outputs: weight, systolic BP, fasting glucose, resting HR and a 0-100 risk score
- Transparent, explainable model: every coefficient is readable and editable
- Delivered as a Streamlit dashboard and a CLI demo, with unit tests

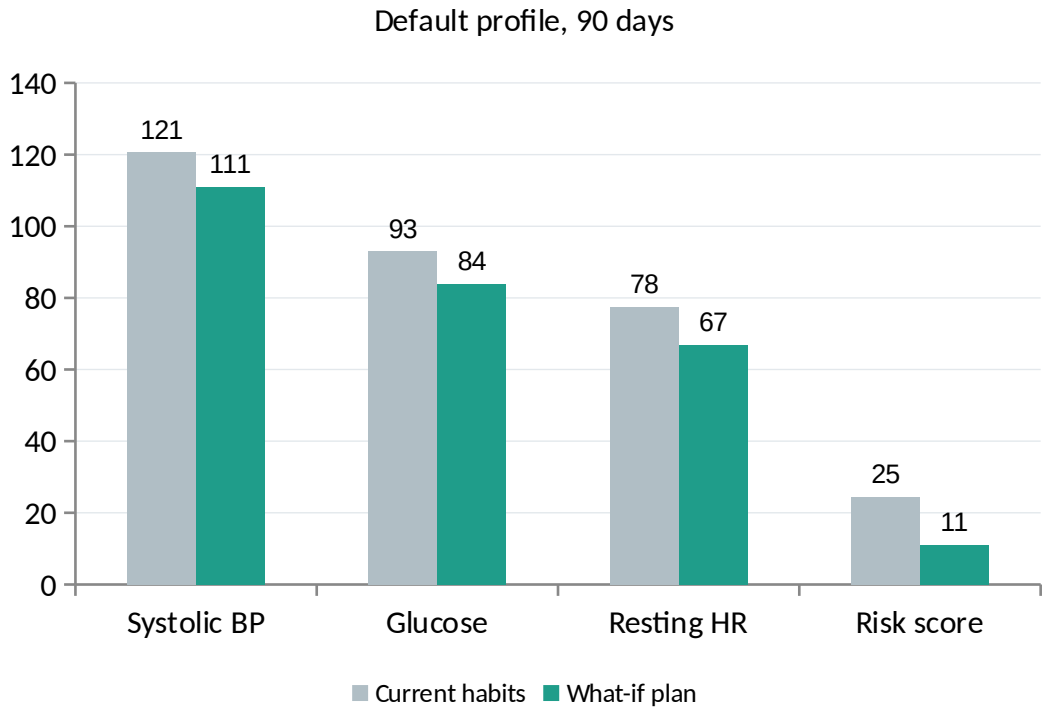
Architecture



Model and AI/ML Framework

- Mechanistic state-space model: weight from energy balance (Mifflin-St Jeor BMR + activity, 7700 kcal/kg)
- BP, glucose and heart rate relax toward lifestyle-dependent targets (30-day time constant)
- Composite risk score from banded biomarkers (BP, glucose, BMI, HR, age)
- Stack: Python, NumPy, pandas, Streamlit, pytest
- Honest status: coefficients are illustrative, not yet clinically validated; ML calibration is on the roadmap

Outcome: Day-90 Simulation



Healthier plan vs current habits
9,000 steps, 7.5 h sleep, 2,200 kcal, 2.3 g sodium, low stress

Weight -5.2 kg
Systolic BP -9.5 mmHg
Glucose -9.0 mg/dL
Risk score -13.4 points

Roadmap, Privacy and Impact

- Calibrate to real wearable and lab data per user; add uncertainty bands (Monte Carlo)
- Learn target functions from public datasets; clinician and coaching views
- Privacy by design: no PII stored, computation local, consent-first data connection
- Educational prototype; not a medical device and not medical advice
- Impact: earlier, more motivated prevention at low cost